

William Hicklin

Systems Architect · Rust Developer · Embedded Engineer

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Professional Summary

Results-driven systems engineer with a decade of experience developing secure, reliable software for connected devices. Proficient in Rust, C++ and embedded Linux, delivering high-quality embedded solutions through cross-functional collaboration and technical leadership. Passionate about advancing memory-safe programming practices in embedded systems through Rust adoption and open-source contributions.

Experience

Embedded Systems Architect

Bristol, UK

Sensoteq

Jan 2026 - Present

A fixed-term remote contract helping Sensoteq design and develop an ultrasonic data logger. Achievements include

- The greenfield design and development of a multi-threaded Rust application collecting and processing ultra-sonic data from multiple co-processors.
- Architecting the communications protocol across the entire system, from embedded IPC to cloud applications, based on Cap'n Proto.
- Developing a version controlled specification document detailing the behaviour and communication APIs for all components complete with state machines, sequence diagrams and normative prose.
- Designing and implementing code patterns enforcing spec-defined API constraints and enabling code reuse across Go and Rust.
- Introducing CI workflows using GitHub Actions covering Cap'n Proto code generation validation, unit testing, and static analysis for Go and Rust, enforcing exhaustive error handling and branch coverage.
- Introducing Nix flake development environments, now adopted across the team and CI pipeline, ensuring identical reproducible toolchains between local development and automated builds.
- Mentoring engineers through detailed PR reviews and ad-hoc conversations.
- Debugging hardware prototypes across the signal and firmware boundary using oscilloscopes and logic analysers, unblocking engineers on timing and signal integrity issues.

Rust software engineer

Bristol, UK

Open-source

Aug 2025 - Jan 2026

Working on the Matter Smart Home Standard Rust SDK as an open-source contributor. I engage with tech leads from a number of organisations including Google, Espressif and the CSA and carry out the following tasks.

- Lead the design and implementation of application level APIs, otherwise know as “clusters”.
- Add certification tests in CI against implemented clusters.
- Produce documentation for SDK users and contributors.
- Develop real-world examples and demonstrations.
- Present work to the community.

Lead Embedded Engineer

Bristol, UK

Dyson

Apr 2024 - Oct 2025

Leading embedded software development for connected devices, focusing on Matter protocol implementation and Rust adoption.

- **Matter Protocol Development:** Delivered the Service Area Cluster for Matter 1.4 specification and SDK, meeting all deadlines while producing comprehensive specifications, test plans, and implementation code.
- **Process Improvement:** Provided comprehensive feedback on the Matter development workflow, identifying pain points and proposing actionable solutions to technical leadership.
- **Open-Source Initiative:** Secured official approval for open-source software contributions and established new internal workflows, creating practical guides for engineers to contribute to open-source projects.
- **Rust Adoption Strategy:** Initiated efforts to promote Rust across the organisation, developed core libraries, created proof-of-concept examples, and contributed to open-source embedded Rust projects including embassy-rs/trouble and rust-embedded/heapless.

Senior Embedded Software Engineer

Bristol, UK

Dyson

Feb 2020 - Apr 2024

Worked on a rapid prototyping team, delivering innovative prototypes and building platforms to accelerate development.

- Led Matter standard implementation efforts, collaborating with engineers across organisations to define specifications and implement new clusters for various device types.
- Designed and implemented embedded hardware and software solutions, including PCBs, back-end software, and embedded UIs.
- Built and maintained a custom OS using Yocto framework, creating tools for internal and external teams.
- Mentored junior team members and led technical aspects of various projects.
- Developed containerised applications and drivers for rapid prototyping platform.

Systems Developer

Air Monitors Limited

Tewkesbury, UK

Feb 2018 - Jan 2020

Led the development of an industrial embedded Linux data-logger for collecting data from air quality instruments.

- Designed and built hardware, developed testing applications, and implemented core software in C++.
- Created a custom OS with cross-compilers and an update server, established cloud infrastructure, and developed a web interface.
- Introduced and managed DevOps pipelines with comprehensive feature unit tests.
- Contributed to the Breathe London project with Google and Imperial College London, responsible for logging air quality and position data from mobile monitoring units.

Research Support Officer

University of Malta

Msida, Malta

Jan 2015 - Feb 2018

Managed a Mobile Air Quality Lab, building systems to manage instruments and data while conducting research campaigns.

- Built online systems to manage instruments and their data using Python and Django.
- Created standard operating procedures for laboratory equipment.
- Performed equipment maintenance and repair.
- Designed and conducted measurement campaigns for air quality research.

Systems Developer (Contract)

University of Malta

Msida, Malta

May 2014 - Nov 2015

Developed an energy-efficient data logger for a seismometer for an international ocean-bottom station project.

- Designed and implemented an embedded system for seismic data collection.
- Collaborated with international team of engineers in Palermo to integrate with the primary system.
- Optimised power consumption for long-term underwater deployment.

Skills

Spoken Languages

English, Maltese

Languages

Rust, C++, Go, Python

Systems

Embedded Linux, Matter, Nix, Balena, Embedded Rust, Yocto

Tools

Altium Designer, PlantUML

Professional Development

Comprehensive Rust Training

Ferrous Systems

Jan 2024

Advanced C Programming

Feabhas

AC-401

Feb 2022

Developing Embedded Systems with Yocto

DOULOS

Feb 2019

Education

University of Malta

M.Sc. in Physics (Research)

Msida, Malta

2012 - 2014

- Grade: Distinction
- Dissertation: "Investigations of VOCs in buildings close to service stations"

University of Malta

B.Sc. (Honours) in Chemistry and Physics

Msida, Malta

2008 - 2012

- Grade: Merit (Second Upper)

- Dissertation: “Construction and Characterisation of Dye-Sensitised Solar Cells: using natural dyes”

Professional Strengths

- Technical leadership and mentorship
- Cross-functional collaboration
- Detailed code review and documentation
- Problem-solving and innovation
- Continuous improvement mindset
- Open-source contribution and advocacy
- Strong analytical thinking and attention to detail